1. Motivation & Model Type Selection

Electoral volatility determines where campaign resources have the highest return on investment. We modeled county-level electoral volatility using demographic features from ACS data merged with election outcomes across Pennsylvania, Michigan, and North Carolina. Our workflow included preprocessing, classification, regression, and clustering models to test whether county demographics can explain or predict swing behavior. We compared models using metrics appropriate to each task.

2. Data Preprocessing

2.1 Data Formatting For Models

The following transformations were completed during Milestone 2 to prepare for modeling. To recap:

- **Log Transformation & Scaling:** Log and StandardScaler were applied to highly right-skewed variables to aid interpretability of linear regression models and comparability of distances in K-means clustering; applied to all models for consistency.
    - Applied to*: total_population, population_density, median_household_income, median_home_value, median_gross_rent*
- **Reducing Multicollinearity:** Required for SVM and logistic/linear regression stability and interpretability; applied to all models for consistency.
    - Dropped fields*: pct_renter_occupied, pct_no_hs_diploma, pct_white, land_area_sqmi*

Feature Examples: Before and After Preprocessing

Stage	total_population	pct_below_poverty
Raw	mean=126,437; range [2,144–1,603,797]	mean=14.8%; range [4.1%–35.2%]
After log	mean=10.8; range [7.7–14.3]	(unchanged)
After Standard Scale	mean=0.0; range [−2.6–2.9]	mean=0.0; range [−1.8–3.3]

Table 1: Pre and post data formatting examples for total_population and pct_below_poverty

2.2 Target Engineering

Our target *vol_z_abs_sum* measures total swing across two election cycles: the sum of absolute z-scored margin changes from 2016 to 2020 and from 2020 to 2024. For classification purposes, the continuous variable was binned into 5-class quantiles (50 each) and binary: High (Q4, Q5, n=100) vs Low (Q1, Q2, Q3, n=150).

5-class Volatility Bins

Quantile Title	Median Volatility Score	Count
Highly Stable	-1.674500	50
Stable	-0.954200	50
Moderate	-0.314400	50
Volatile	0.607100	50
Highly Volatile	2.076000	50

Table 2: 5-class target variable quantiles

2.3 Feature Selection

28 demographic variables were selected for classification and cluster modeling. All election variables were excluded to prevent leakage. The six thematic groups include:

Group	Features	n
Race / Ethnicity	*pct_black, pct_asian, pct_two_or_more_races, pct_hispanic, pct_non_hispanic_white, race_entropy_norm*	6
Urbanization	*log_total_population, log_population_density, pct_drive_alone, pct_carpool, pct_public_transit, pct_work_from_home*	6
Education	*pct_hs_or_higher, pct_bachelors_plus*	2
Housing	*log_median_gross_rent, log_median_home_value, pct_owner_occupied*	3
Economic	*log_median_household_income, pct_below_poverty, pct_income_under_25k, pct_income_50k_100k, unemployment_rate*	5
Household / Age	*median_age, pct_senior_65plus, pct_young_adult_18_24,*	7

	pct_foreign_born, pct_family_households, pct_married_couple, pct_living_alone	

Table 3: Thematic organization of demographic attribution for classification and cluster analysis

3. Classification Models

Campaign strategists need to know: *"Could this county swing significantly?"* This naturally invokes a categorization task to identify electoral opportunities. Our goal is to use only knowable, non-electoral demographic data to identify counties that may represent future opportunities for Democratic campaigns.

We compared classifiers from different algorithmic families, including ensemble trees, gradient boosting, and kernel methods. The variety of classification techniques helps assess predictive performance while also revealing whether the relationship between demographics and volatility is primarily linear or nonlinear.

Model Selection Summary

Model	Requires Numerical?	Requires Scaling?	Our Approach
Random Forest	No	No	Scaling for consistency
XGBoost	No	No	Labels 0-indexed
SVM	Yes	Yes	StandardScaler required

Table 4: Classification models, requirements, and approach

3.1 Random Forest

3.1.1 Rationale

Random Forest is nonparametric and does not require distribution assumptions among predictors. This is applicable in our dataset as electoral volatility likely reflects interactions between socioeconomic and demographic variables rather than simple linear combinations. Further, Random Forest models are more robust to overfitting/noise relative to simple decision trees. Lastly, it provides an interpretable feature importance framework.

3.1.2 Assumptions

The primary practical assumption is that the predictors contain a meaningful signal related to electoral volatility. Due to its non-parametric nature, we do not need to make assumptions regarding linearity, normality, or homoskedasticity.

3.1.3 Hyperparameters

Inner 3-fold cross-validation for hyperparameter tuning.

Setting	Search Space	Best	Controls
n_estimators	100–500	300	Number of trees
max_depth	5–20, None	8	Tree complexity
min_samples_leaf	3, 5, 10	5	Prevents memorization
max_features	sqrt, log2, 0.3, 0.5	sqrt	Diversity among trees
class_weight	balanced, subsample, none	balanced	Handles class imbalance

Table 5: Random Forest hyperparameters, tuning ranges, selected values, and effect on model behavior

3.1.4 Performance Evaluation

Random Forest performed well above the random-guess baseline (20%) on the 5-level volatility classification, achieving 36.8% overall accuracy and 0.362 macro F1. Performance peaked at the extremes of the volatility spectrum with F1 scores of 0.59 and 0.46 for Highly Volatile and Highly Stable, respectively. The model struggled with middle classes, as the confusion matrix illustrates strong diagonals for Q1 and Q5 but diffuse misclassification among Q2, Q3, and Q4. Q2 was correctly classified at the approximate random-guess baseline.

Outer 5-fold cross validation for model evaluation.

Metric	Value
Accuracy	0.368
Precision (macro)	0.369
Recall (macro)	0.368
F1-macro	0.362
ROC-AUC	0.803*

*Table 6: 5-class Random Forest classification results; * indicates best overall*

F1 Comparison

Volatility Class	Highly Stable	Stable	Moderate	Volatile	Highly Volatile
F1	0.46	0.18	0.28	0.30	0.59

Table 7: Random Forest F1 score distribution across volatility classes

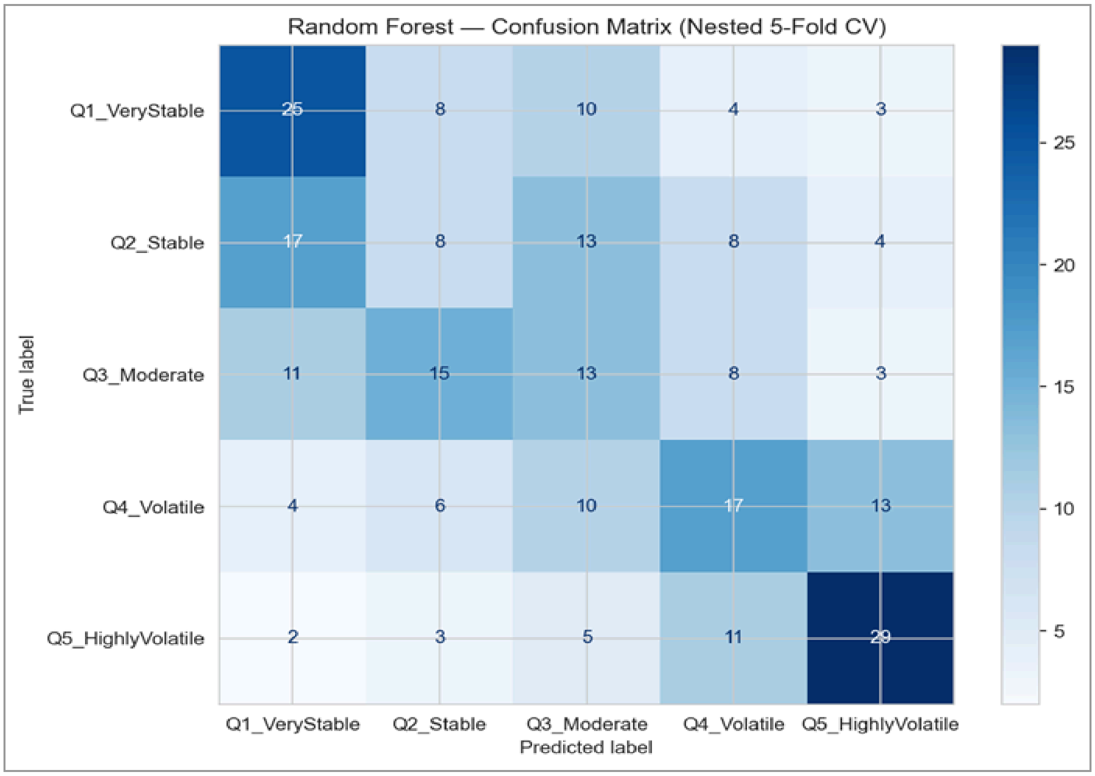

Figure 1: Random Forest confusion matrix with strong diagonal at Q1 and Q5; misclassification among middle classes

Feature importance was evaluated using permutation importance (model-agnostic, 30 repeats) rather than standard Gini importance, as permutation-based rankings are less biased toward continuous predictors. Permutation importance was computed using F1-macro as the scoring metric, so feature importance reflects the decline in macro F1 after shuffling each variable. *race_entropy_norm* was identified as the most important predictor in the Random Forest model, as shuffling reduced macro F1 by 0.042 on average.

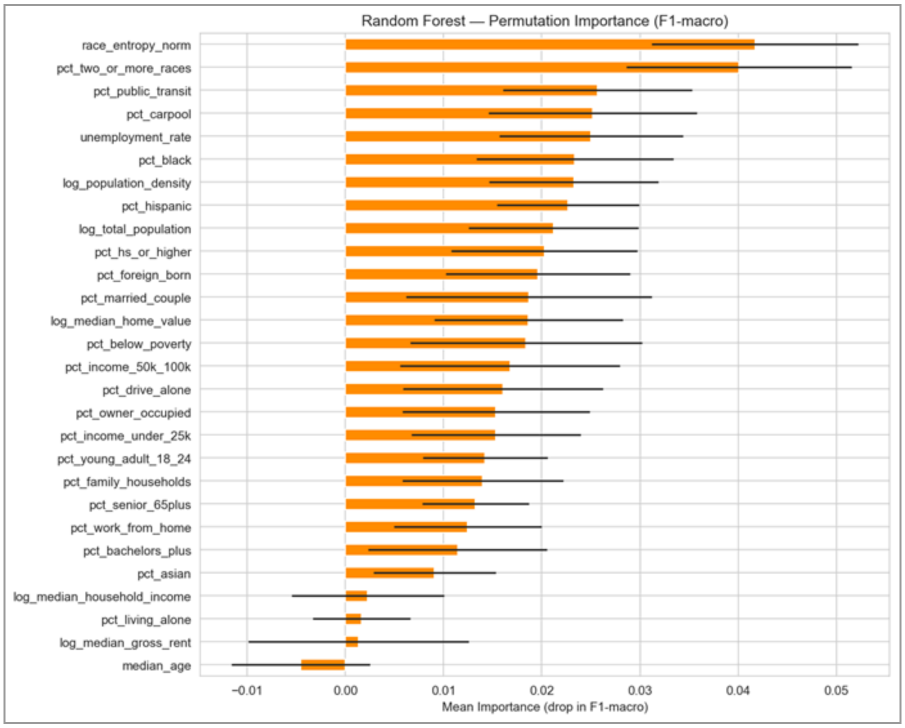

Figure 2: Permutation importance — race_entropy_norm (diversity index) is the single most important feature

3.2 XGBoost

3.2.1 Rationale

Like Random Forest, XGBoost is a nonparametric tree-based method; it does not assume linear relationships between predictors and the response and can capture nonlinear patterns and interactions in tabular data. XGBoost's most notable advantage over Random Forest is in its sequential error correction. This was particularly relevant within the middle volatility classes, where blurred boundaries were more difficult to distinguish than the extremes. Unlike permutation importance, SHAP provides county-level explanations that show both the direction of each feature's effect and its contribution to specific class predictions.

3.2.2 Assumptions

Because XGBoost builds trees sequentially, it assumes that remaining prediction errors contain information or patterns that later trees can improve upon. Due to its non-parametric nature, we do not need to make assumptions regarding linearity, normality, or homoskedasticity.

3.2.3 Hyperparameters

Inner 3-fold cross-validation for hyperparameter tuning.

Setting	Search Space	Best	Controls
n_estimators	100–500	300	Total learning capacity
max_depth	3–8	5	Complexity per tree
learning_rate	0.01–0.2	0.1	Step size per tree
subsample	0.6–1.0	0.8	Fraction of data per tree
colsample_bytree	0.6–1.0	0.8	Fraction of features per tree
reg_alpha (L1)	0–1	0.1	Feature selection penalty
reg_lambda (L2)	1–10	5	Prediction smoothing

Table 8: XGBoost hyperparameters, tuning ranges, selected values, and effect on model behavior

3.2.4 Performance Evaluation

XGBoost slightly outpaced Random Forest's overall performance metrics, achieving a 38.8% accuracy and 0.387 macro F1. Like Random Forest, XGBoost performed best at the extreme volatility classes; however, the model's clearest advantage was its ability to distinguish the middle volatility classes as shown in the F1 comparison. Despite improvements throughout the middle classes, Q2 classification remained in line with the random-guess baseline.

Outer 5-fold cross validation for model evaluation.

Metric	Value
Accuracy	0.388*
Precision (macro)	0.396
Recall (macro)	0.383
F1-macro	0.387*
ROC-AUC	0.783

*Table 9: 5-class XGBoost classification performance summary; * indicates best overall*

F1 Comparison

Volatility Class	Highly Stable	Stable	Moderate	Volatile	Highly Volatile
F1	0.37	0.27	0.33	0.39	0.56

Table 10: XGBoost F1 score distribution across volatility classes

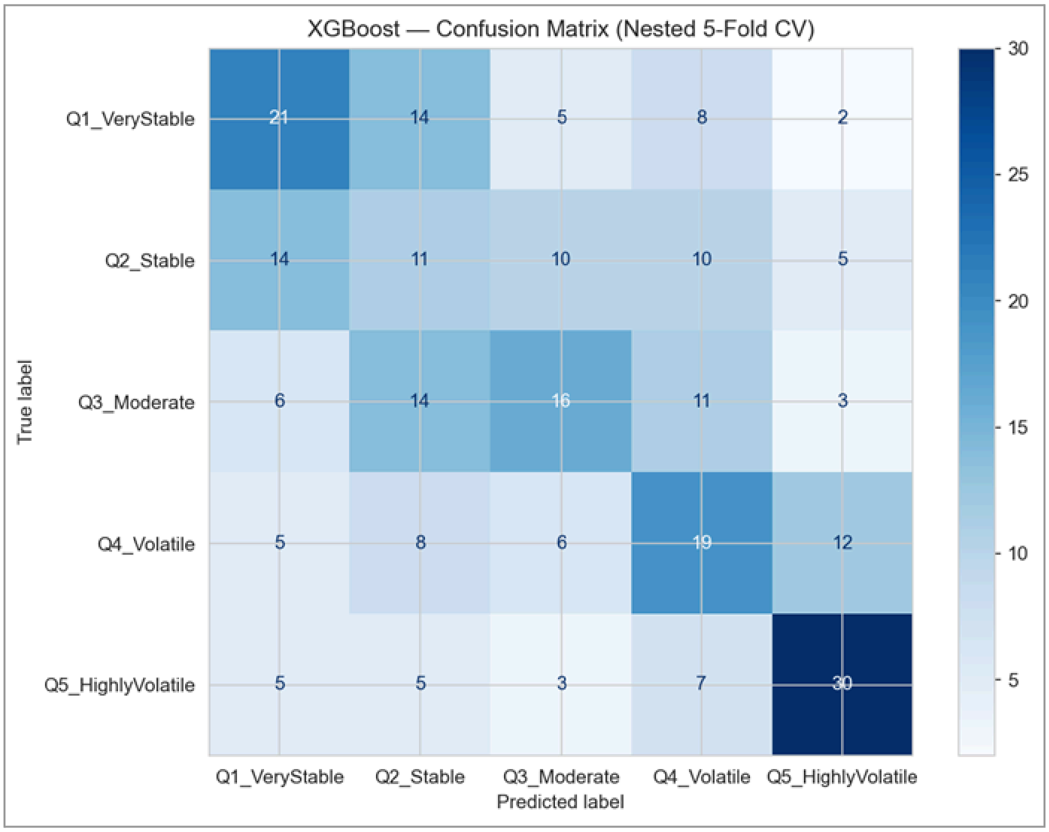

Figure 3: XGBoost confusion matrix with stronger diagonal at Q1 and Q5 with improved classification among middle classes

SHapley Additive exPlanations (SHAP) was employed to understand which demographics were the most impactful in driving the XGBoost volatility predictions. Housing costs (*log_median_gross_rent*) was the dominant predictor of class despite having a negligible impact on F1-macro in the Random Forest permutation importance analysis. The variable was particularly determinative for the Highest Volatility class (Q5). Similar to Random Forest, racial diversity (*race_entropy_norm*) illustrated the second-largest impact in the SHAP analysis.

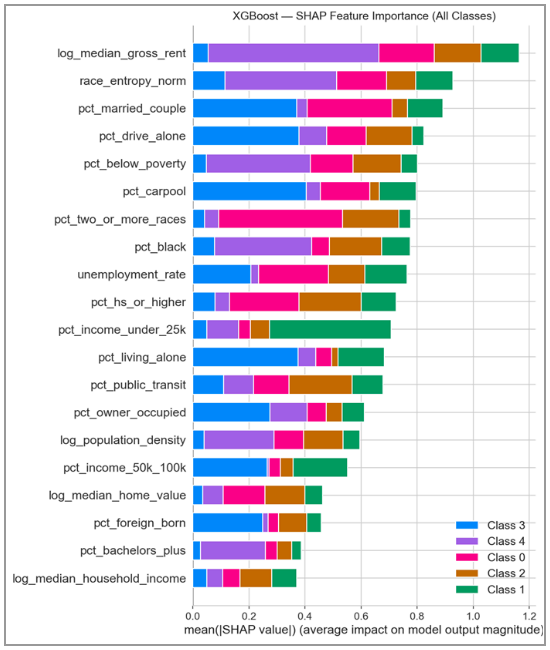

Figure 4: XGBoost SHAP analysis; color defines class impact; housing cost and racial entropy among highest importance

In addition to magnitude, SHAP dependence plots show the direction of each predictor's contribution to the model output, particularly on the Highly Volatile classification. Log median gross rent and race entropy both illustrate a direct relationship with volatility, indicating that higher values of each pushed counties toward the Highly Volatile class. Percent below poverty showed a threshold-like relationship, where only higher values sharply increased the model's Q5 prediction.

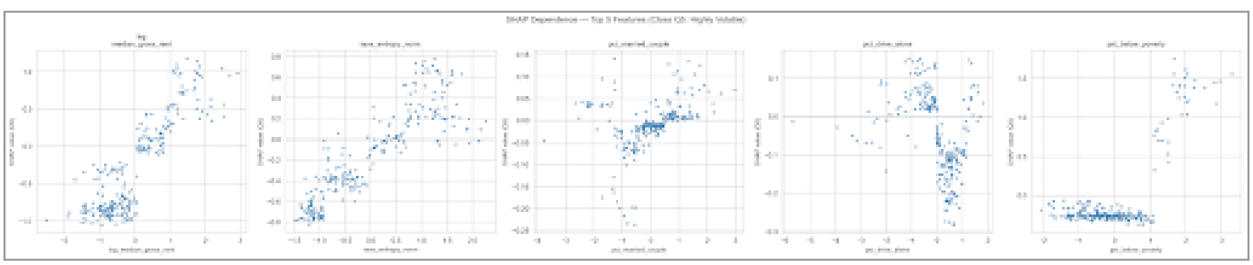

Figure 5: Dependence plots for top 5 SHAP importance features

3.3 SVM (Support Vector Machine)

3.3.1 Rationale

SVM's interchangeable kernels allowed us to test whether the volatility classes were separated by approximately straight partitions or nonlinear boundaries. Therefore, SVM served as a diagnostic tool in addition to its utility as a predictive model.

3.3.2 Assumptions

The assumptions required for SVM depend on kernel selection. The linear kernel assumes class boundaries are approximately straight. The RBF kernel assumes nonlinear boundaries and that counties with similar features will behave similarly. Because the model is reliant on maximum-margin boundaries, SVM assumes a scaled dataset.

3.3.3 Hyperparameters

Inner 3-fold cross-validation for hyperparameter tuning.

Kernel	C	Gamma	class_weight	Selected C
Linear	0.01–100	N/A	balanced	1
RBF	0.01–100	scale, auto, 0.01–1	balanced	10

Table 11: SVM hyperparameter tuning ranges and selected values for the linear and RBF kernels

3.3.4 Performance Evaluation

Outer 5-fold cross validation for model evaluation.

SVM: Kernel Comparison

Metric	Linear	RBF
Accuracy	0.368	0.340
F1-macro	0.364	0.347
ROC-AUC	0.765	0.741
Precision-macro	0.365	0.346
Recall-macro	0.368	0.352

Table 12: SVM classification kernel comparison and performance summary

In cross validation, the linear kernel illustrated stronger results relative to RBF across all five classification performance metrics. The 5 volatility levels appear to be separated more effectively by linear boundaries.

F1 Comparison (Linear)

Volatility Class	Highly Stable	Stable	Moderate	Volatile	Highly Volatile
F1	0.456	0.365	0.191	0.344	0.463

Table 13: SVM F1 score distribution across volatility classes

3.4 Classification Model Comparison

Model	Accuracy	F1-macro	ROC-AUC	Best At
XGBoost	0.388	0.387	0.783	Best overall
SVM (Linear)	0.368	0.364	0.765	Middle classes
Random Forest	0.368	0.362	0.803	Best ROC-AUC

Table 14: Performance metric comparison across classification models

XGBoost was the strongest overall classifier, achieving the highest macro F1 and accuracy across the full five-class target. Its advantage was most apparent in the middle volatility classes, where county profiles were less clearly separated and boosting helped improve difficult boundary cases with sequential error correction.

From a campaign-targeting perspective, however, Random Forest may be the most useful model because it achieved the highest F1 for the Highly Volatile class. Since Highly Volatile counties are the most actionable group in practice, class-specific performance for that category is more important than overall macro F1 alone. In other words, XGBoost performed best in the broad, context-agnostic sense of overall multiclass classification, while Random Forest performed best for the project's specific decision-making objective.

3.5 Classification Deep Dive

3.5.1 State-Level Classification Extension

The three-state pooled models in Sections 3.1 through 3.3 treat all 250 counties as a single population. A natural follow-up question is whether the demographic predictors that drive volatility generalize across state lines or whether each state contains its own distinct volatility mechanism. To test this, we conducted a "Leave-One-State-Out" (LOSO) experiment using the best-performing 5-class model, XGBoost. In each rotation, two states served as the training set and the third as the held-out test set. The StandardScaler was re-fit on the training states to prevent any information leakage from the test state into the feature scaling step. If the demographic signal generalizes, cross-state accuracy should remain well above the random-guess baseline of 20%.

Test State	Train States	Accuracy	F1-macro
Michigan	PA + NC	0.205	0.182
North Carolina	PA + MI	0.210	0.213
Pennsylvania	MI + NC	0.209	0.189

Table 15: Leave-one-state out classification performance metrics

All three LOSO rotations produced accuracy and F1 scores that are effectively at random-guess baseline, 0.200. The average LOSO F1-macro was 0.195, compared to 0.387 for the pooled 5-fold cross-validation model – a collapse in predictive power. The confusion matrices in Figure 7 confirm this finding: predictions are spread diffusely across all five classes rather than concentrated along the diagonal. None of the three rotations exhibits a systematic pattern of misclassification; the model simply cannot distinguish the volatility classes when applied to a state it has not been trained on.

To understand why cross-state prediction fails, we trained separate within-state binary classifiers (High vs Low Volatility) using XGBoost with 5-fold cross-validation and extracted feature importance rankings for each state independently. The results reveal that the demographic drivers of volatility differ substantially across the three states.

Rank	PA (Diversity)	MI (Education)	NC (Poverty)
#1	*race_entropy_norm*	*pct_bachelors_plus*	*pct_below_poverty*
#2	*log_median_gross_rent*	*log_median_gross_rent*	*log_median_gross_rent*
#3	*pct_foreign_born*	*pct_work_from_home*	*log_population_density*

Table 16: The top 3 drivers of volatility for within-state models

In Pennsylvania, racial diversity (*race_entropy_norm*) dominates, with housing cost and foreign-born population as secondary drivers. Michigan's volatility is driven by education attainment (*pct_bachelors_plus*), followed by housing cost and work-from-home prevalence. North Carolina's volatility is best predicted by poverty rate (*pct_below_poverty*), with housing cost and population density in supporting roles. The divergence in top predictors may reflect distinct state-level political economies.

The one feature that appears among the top predictors in all three states is housing cost (*log_median_gross_rent*). This suggests that housing cost may serve as a proxy for economic transition - a signal that is relevant regardless of local politics. Figure 6 illustrates these per-state importance rankings side by side.

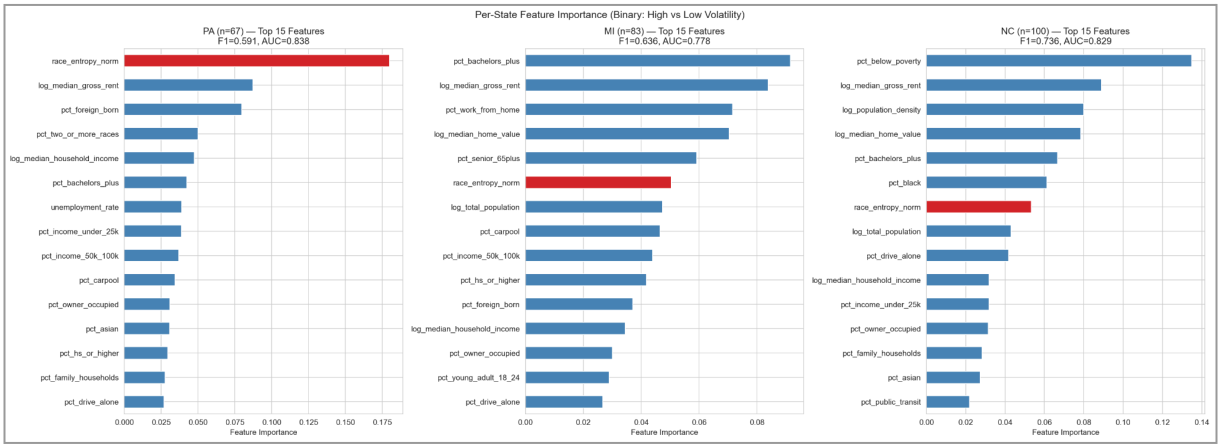

Figure 6: Per-state feature importance

Figure 7 displays the LOSO confusion matrices for all three rotations. There is no visible diagonal structure, confirming that the cross-state model has no predictive value. This contrasts with the pooled confusion matrices in Sections 3.1 and 3.2, where the extremes (Q1 and Q5) showed meaningful concentration along the diagonal.

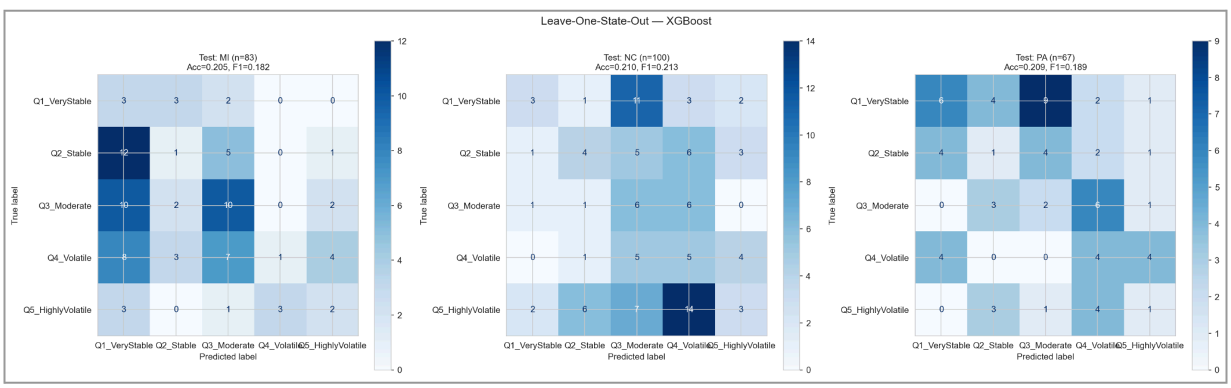

Figure 7: LOSO confusion matrices. All three rotations = near-random.

Although cross-state transfer fails, within-state models perform substantially better. We fit binary XGBoost classifiers (High vs Low Volatility) within each state separately using 5-fold cross-validation. Within-state scaling was applied independently for each state to prevent any data leakage.

State	n	Within-State AUC	Within-State F1
PA	67	0.838	0.591
MI	83	0.778	0.636
NC	100	0.829	0.736

Table 17: Within-state AUC and F1 classification metrics

All three state-specific models achieve AUC values between 0.778 and 0.838, indicating strong discriminative ability when trained and tested within the same state. North Carolina achieves the highest within-state F1 (0.736), suggesting that its volatility is most tightly coupled to demographic indicators. This is consistent with the poverty-driven mechanism identified in the feature importance analysis: NC counties with high poverty rates reliably fall into the high-volatility group, providing a cleaner signal for the classifier. Michigan and Pennsylvania also achieve respectable performance (F1 of 0.636 and 0.591, respectively), though the smaller sample sizes and more heterogeneous volatility drivers introduce additional noise.

The LOSO experiment yields a clear conclusion: demographics predict volatility well within individual states (AUC 0.78 to 0.84) but fail entirely across state lines (F1 approximately equal to random guessing at 0.195). The performance discrepancies by model scope suggest that the relationship between demographics and electoral volatility is mediated by state-specific political cultures, historical voting patterns, and local economic structures that cannot be captured by census variables alone. State-specific models, trained on local data, are necessary to capture the distinct mechanisms at work in each battleground state. Housing cost (*log_median_gross_rent*) is the only predictor that generalizes across all three states, making it the single most robust indicator for cross-state volatility screening.

3.5.2 Binary Classifier

The 5-class classification models in Sections 3.1 through 3.3 achieved F1-macro scores in the range of 0.36 to 0.39, with performance weakest in the middle volatility classes where county demographic profiles overlap substantially. As a follow-up experiment, we tested whether collapsing the target from five quantile classes to a binary split would yield a more reliable classifier. The binary target grouped Q4 and Q5 as High Volatility (n=100) and Q1 through Q3 as Low volatility (n=150). All five classification models were retrained on this binary target using the same nested 5-fold cross-validation framework and identical preprocessing pipeline.

Model	Accuracy	F1	Precision	Recall	AUC
Random Forest	0.780	0.718	0.726	0.690	0.828
XGBoost	0.748	0.667	0.718	0.610	0.803
SVM (RBF)	0.760	0.674	0.724	0.630	0.789
Logistic Regression	0.680	0.604	0.598	0.610	0.745
SVM (Linear)	0.688	0.610	0.610	0.610	0.734

Table 18: Binary classification performance summary by model

Random Forest achieved the strongest binary performance across all five metrics, with an F1 of 0.718, accuracy of 0.780, and AUC of 0.828. XGBoost and SVM (RBF) followed closely, while the linear models (Logistic Regression and SVM Linear) lagged behind. Notably, the model ranking shifted relative to the five-class task: Random Forest, which placed third in macro F1 on the 5-class problem, became the clear leader in the binary setting. This is consistent with its earlier strength on the Highly Volatile class, where it achieved the highest per-class F1 of 0.59.

Performance was substantially stronger in the binary setting than in the five-class setting as F1 nearly doubled from 0.387 to 0.718. The F1 gain reflects the elimination of the ambiguous middle classes that accounted for most of the misclassification error in the 5-class task. However, it is important to consider the simpler binary prediction task and higher majority-class baseline (60%). AUC improved modestly from 0.803 to 0.828, suggesting that the models were already ranking counties reasonably well by volatility risk; the binary restructuring improved their ability to convert the ranking into the correct class allocation.

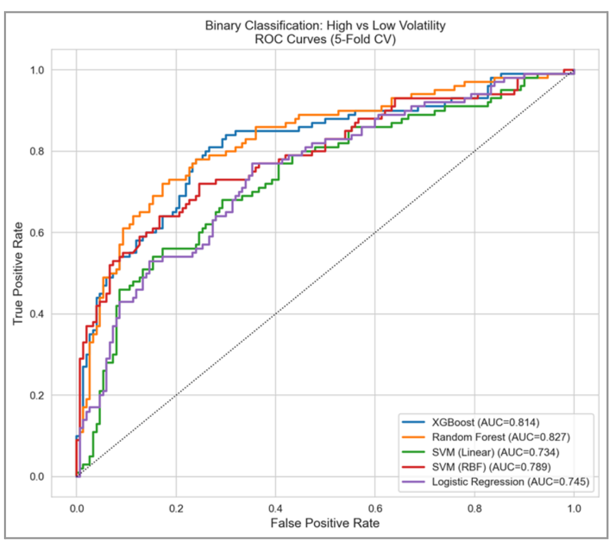

Figure 8: Binary ROC curves

Figure 8 displays the binary ROC curves for all five models. Random Forest maintains the highest true positive rate across nearly all false positive rate thresholds. The separation between the tree-based models (Random Forest, XGBoost) and the linear models (Logistic Regression, SVM Linear) is visible throughout the mid-range of the curve, reinforcing that the High vs Low Volatility benefits from nonlinear modeling. SVM with the RBF kernel occupies an intermediate position, outperforming the linear models but falls short of Random Forest.

3.6 Challenges & Limitations

Across the 3 classification models, overall 5-class performance remained modest, particularly for the middle volatility groups where county profiles blur. This likely reflects both the structure of the target variable and limitations in the available predictors.

County-level electoral volatility is probably influenced by factors not captured in the demographic dataset alone, such as local industry composition, migration patterns, campaign-specific dynamics, media environment, among other politically adjacent variables that may be more difficult to quantify. Structurally, it is not surprising that the middle volatility classes were the most difficult to distinguish. Counties in these categories have volatility scores that are closer together than counties at the extremes, so the class boundaries are naturally less distinct. Counties on the margins of the distribution are more clearly separated in their underlying volatility values, making them easier to distinguish. The models performed substantially better on the Highly Volatile class, which is the most actionable category in practice. The binary classification results indicate that demographics retain practical predictive value when the problem is framed around high-versus-low volatility rather than granular class separation.

Macro F1 was used for overall model comparison because it balances precision and recall across all classes. In application, however, precision for the Highly Volatile class is of particular importance because false positives could lead to inefficient allocation of limited campaign resources. Further, classification requires discrete outcomes, whereas our volatility response variable is continuous. Our solution involved the artificial binning of counties into 5 classes. Next, we address the limitation with linear regression, modelling the continuous volatility measure more precisely.

Lastly, the results indicate a limitation in scalability. The increased predictive power of the within-state models paired with the near-random performance in the LOSO setting suggests a single pooled classification model is unlikely to generalize well across a national landscape.

4. Regression Model

We used multiple linear regression to predict the continuous volatility score, *vol_z_abs_sum*, from demographic county characteristics. Unlike the classification models, which assigned counties to one of five discrete volatility classes, regression modeled the continuous numeric volatility measure directly. This preserved more information than the binned outcome and allowed us to estimate how strongly each demographic feature was associated with changes in volatility, rather than only predicting a class label.

4.1 Multiple Linear Regression

4.1.1 Rationale

Multiple linear regression provided an interpretable model for understanding how demographic variables relate to volatility. Unlike the classification models, which focused on discrete class assignment, regression estimated average predictor-response relationships directly on the continuous outcome. The model provided an inherent coefficient-based measure of predictor importance, including both direction and magnitude.

4.1.2 Assumptions

The model assumes linear relationships between predictors and the response, constant variance of residuals, independent observations, approximately normal residuals, and limited multicollinearity among predictors.

4.1.3 Model Specification / Assumption Checks

We tried four different approaches to feature selection for the final model: forward selection with R^2 scoring, forward selection with mean absolute error scoring, backward selection with R^2 scoring, and backward selection with mean absolute error scoring. Each approach was applied to the same randomly selected 80% training set using 5-fold cross-validation. We then calculated the AIC and BIC scores of the models using the feature subsets returned by these techniques and selected the model that provided the best tradeoff across AIC and BIC.

Selection Technique	Scoring	AIC + BIC
Forward Select	R2	1,360.2
Forward Select	Mean Abs Error	1,351.4
Backward Select	R2	1,352.2
Backward Select	Mean Abs Error	1,351.4

Table 19: Comparison of feature-selection approaches for the pooled multiple linear regression model

The two MAE-based selection procedures produced the lowest combined AIC and BIC score

(660.9 + 690.5 = 1351.4). This model retained percent Black, percent bachelor's degree or higher, percent income under $25k, percent owner-occupied housing, percent drive-alone commuting, racial entropy, log median home value, and log median gross rent.

Selected Predictors	**Coefficient**
pct_black	0.2997
pct_bachelors_plus	0.5816
pct_income_under25k	0.6622
pct_owner_occupied	0.5474
pct_drive_alone	0.2343
race_entropy_norm	0.4730
log_median_home_value	-0.6088
log_median_gross_rent	0.8789

Table 20: 3-state MLR model predictors and coefficients

Regarding model assumptions, variance inflation factors were generally acceptable, with most predictors below 5 and only *log_median_gross_rent* slightly above that threshold at 5.34. Diagnostic plots suggested that linearity and residual normality were reasonable with mild deviations in the tails. The scale-location plot indicated slight heteroscedasticity, but not severe enough to invalidate interpretation. The residuals-versus-leverage plot did not reveal any highly influential observations, and county-level observations were treated as independent.

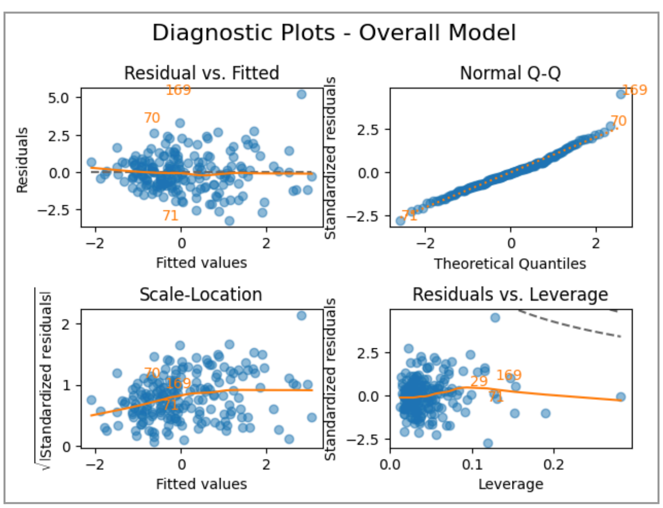

Figure 9: Diagnostic plots for 3-state multiple linear regression model (final model)

4.1.4 Performance Evaluation

On the held-out 20% test set, the model achieved an RMSE of 1.122 and an R^2 of 0.236. These results suggest that the regression captured some meaningful variation in county volatility, but overall predictive strength was modest.

4.1.6 State-Level Regression Extension

Separate regression models were fit for Michigan, North Carolina, and Pennsylvania to examine whether predictor relationships were geographically consistent. We utilized forward selection with MAE scoring and 5-fold cross-validation to fit multiple linear regression models for each state. As in the pooled model, each state dataset was randomly partitioned into an 80% training set and a 20% held-out test set.

Feature	Overall	Michigan	North Carolina	Pennsylvania
const	0.0378	-0.0429	-0.1206	-0.3046
pct_black	0.2997		0.5586	
pct_bachelors_plus	0.5816	0.4513		
pct_income_under_25k	0.6622			
pct_owner_occupied	0.5474			
pct_drive_alone	0.2343			
race_entropy_norm	0.4730			0.5364
log_median_home_value	-0.6088			
log_median_gross_rent	0.8789			0.6135
pct_carpool		-0.2306	0.1359	
pct_hispanic			-0.3821	
pct_below_poverty			0.8313	
log_median_household_income			0.6463	
median_age				0.4626
unemployment_rate				0.4855

Table 21: Comparison of selected predictors and respective coefficients for overall and state-level models

Predictor selection differed substantially by state, suggesting that the demographic drivers of volatility are not fully uniform across the selected electorate battleground states. Michigan and Pennsylvania achieved modest out-of-sample performance (R^2=0.310 and R^2 = 0.216, respectively), while North Carolina performed worse than a baseline model (R^2=−0.034).

Scoring type	Overall	Michigan	North Carolina	Pennsylvania
RMSE	1.122	0.896	2.329	1.013
R^2	0.236	0.310	-0.034	0.216

Table 22: Comparison of performance metrics by model scope

4.2 Challenges & Limitations

The regression model was useful for interpretation, but its predictive performance was limited. This may reflect remaining nonlinear relationships and omitted-variable effects, since county-level volatility is likely influenced by factors not captured in the dataset, such as local industry composition, immigration patterns, current local government officials, or campaign-specific dynamics. Mild heteroscedasticity also remained after preprocessing.

5. Clustering

Unlike supervised classification, clustering forms groups without volatility labels. It provides a cleaner test of two related questions: (1) do counties naturally organize into distinct demographic structures, and (2) do those structures correspond to electoral volatility? Clusters with varying densities of volatile counties suggest interpretable, cross-state county archetypes for electoral volatility. Degrees of alignment between natural clusters and volatility could validate classification results as true patterns rather than overfitting. A uniform distribution of volatile counties across clusters, on the other hand, suggests volatility is more continuous or less attached to demographic indicators.

5.1 K-Means Clustering

5.1.1 Rationale

K-means clustering provides a simple and interpretable method to identify natural groupings in the county-level demographic data. It allowed us to test whether differences in electoral volatility correspond to generalized demographic archetypes. Rather than predicting labeled outcomes for a defined set of classes, K-means evaluates whether volatility patterns emerge from the underlying structure of the counties themselves. In addition, K-means requires the number of clusters to be specified in advance, allowing for comparison across multiple k-values to assess how closely unsupervised groupings are aligned with the five-class target structure used in classification modeling.

5.1.2 Assumptions

K-means assumes that the predictors contain information conducive to clustering based on distance to centroids. Due to its reliance on Euclidean distance, the predictors must be appropriately scaled and numeric. These assumptions align with the makeup of our county-demographics dataset.

5.1.3 Model Specification / Tuning

We fit K-means models across a range of k-values from 2 to 10 and compared solutions using silhouette score, Davies-Bouldin index, and cluster size balance. To reduce sensitivity to starting values, each K-means solution was fit with *n_init* = 25.

k	Silhouette	Davies-Bouldin	Min Cluster Size	Max Cluster Size
2	0.245046	1.656007	78	172
3	0.21908	1.60607	53	137
4	0.196483	1.537421	19	115
5	0.203704	1.501173	13	120
6	0.167042	1.590744	12	99
7	0.165695	1.620173	12	84
8	0.154793	1.597231	12	73
9	0.156395	1.434816	1	73
10	0.154859	1.421279	1	71

Table 23: K-means performance metrics across range of k values (2-10)

Although the highest silhouette score occurred at k=2, this cluster count was less conducive for comparison with our five-class volatility framework. We therefore emphasized the k=5 solution as a compromise: it aligned with our existing class structure, achieved a relatively low Davies-Bouldin score, showed a small rebound in silhouette relative to k=4 and avoided the one-county clusters that appeared at higher k-values.

5.1.4 Performance Evaluation

We analyzed the distribution of our volatility classes within the k=5 clusters.

Cluster Label	Highly Stable	Stable	Moderate	Volatile	Highly Volatile
1	6	7	14	11	9
2	10	7	11	11	8
3	0	2	1	3	22
4	33	32	23	20	7

5	1	2	1	5	4

Table 24: Comparison of volatility class distribution by cluster (k=5)

Interestingly, Cluster 3 was heavily skewed toward Highly Volatile counties, with 22 of its 28 counties falling in that class. This is consistent with the earlier classification results, where racial diversity and housing cost variables were among the most influential predictors of volatility. Cluster 3 trails only Cluster 5 in mean race entropy and has the lowest median rent of any cluster, suggesting that these same factors also help define an unsupervised, high volatility demographic grouping.

Geographically, Cluster 3 appears to be concentrated entirely in North Carolina. The pattern suggests that the clustering may be capturing a state-specific demographic pattern associated with volatility rather than a cross-state archetype. This geographic inconsistency is in accordance with earlier evidence of distinct predictors by state.

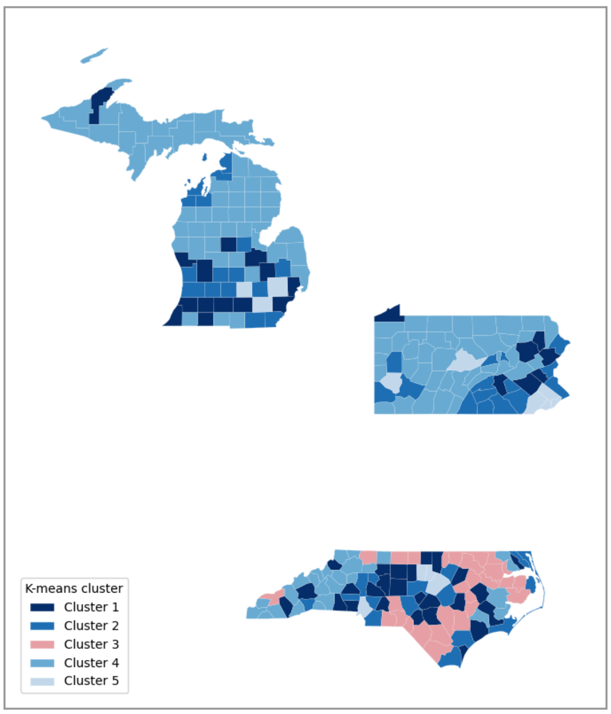

Figure 10: Counties according to their respective cluster (k=5)

5.2 Challenges & Limitations

The clustering results did not illustrate sharply defined groups within the demographic data. The relatively low silhouette scores across all tested k-values suggest substantial overlap between middle volatility counties. This is consistent with the classification analyses, where the middle classes were also the most difficult to distinguish.

In alignment with binary classification results, k=2 produced the strongest internal clustering metrics. In both cases, the data appear to support a broader high-versus-low volatility structure more clearly than a five-class framework, suggesting that electoral volatility may be more naturally organized at a broad level than at a finely partitioned one.

More generally, clustering analyses are not predictive due to their unsupervised nature. Instead, their value in this project serves interpretive and context-building purposes by testing whether natural demographic groupings exist in the data. It is therefore limited in its direct application in addressing the project's primary motive.

6. Further Analysis

While not directly related to our project's motive, we performed a few additional modeling techniques to advance our understanding of the dataset and address natural adjacent questions.

6.1 Logistic Regression

Predict which party a county "should" vote for based on demographics, then analyze the errors. The predicted Democratic probability was obtained from a logistic regression model trained to predict whether a county voted Democratic using demographic variables only. "Misfits" are defined as a county in the top 20% of *misfit_score*:

$$\text{actual\_winner}_i = \begin{cases} 1, & \text{if county } i \text{ voted Democratic} \\ 0, & \text{if county } i \text{ voted Republican} \end{cases}$$

$$\text{misfit\_score}_i = |\text{actual\_winner}_i - P_i(\text{Dem})|$$

6.1.1 Rationale

Predict partisan lean from demographics and find "misfits" - counties where the prediction is confidently incorrect.

6.1.2 Assumptions

Linear boundary between D and R counties; prediction errors capture demographic-political misalignment.

County	State	Model Says	Actually	Misfit	Volatility
Leelanau	MI	96% Rep	Dem +8%	0.96	Highly Volatile
Marquette	MI	96% Rep	Dem +9%	0.96	Highly Volatile
Nash	NC	95% Dem	Rep −2%	0.95	Very Stable
Lenoir	NC	94% Dem	Rep −7%	0.94	Stable
Cabarrus	NC	87% Dem	Rep −8%	0.87	Highly Volatile

Table 25: The top 5 “Misfit” counties in terms of political lean

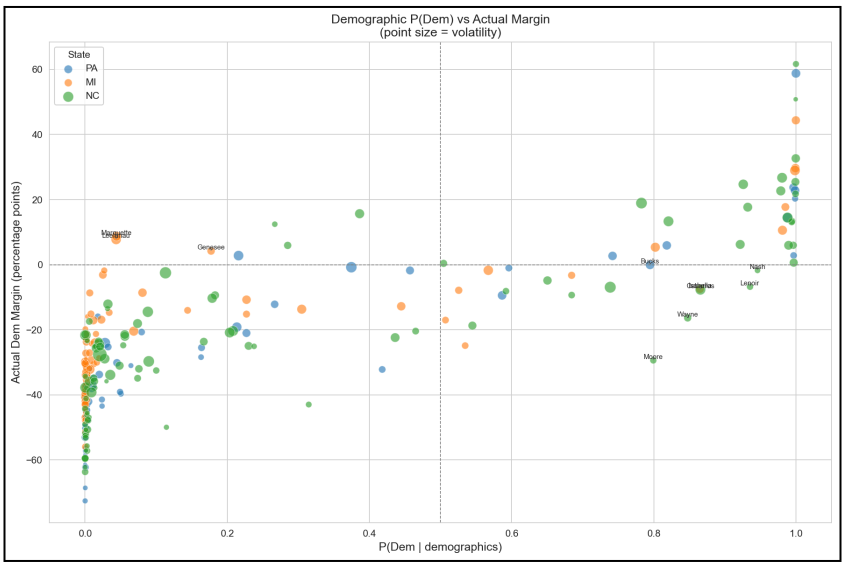

Figure 11: P(Dem) vs actual margin; dot size = volatility; counties far from diagonal are misfits.

Finding	Test	Value
Misfits are more volatile	Spearman ρ	0.408 ($p < 0.001$)
Misfits don't lean one way	Pearson r	−0.017 (n.s.)

Table 26: Statistical significance tests investigating the correlation between “Misfit” metric and volatility

Characteristic	Misfits	Non-Misfits	p-value
Racial diversity	0.567	0.419	< 0.0001
Bachelor's+	30.9%	24.7%	< 0.0001

Homeownership	71.1%	76.3%	< 0.0001
Median rent	$1,056	$932	< 0.0001
Median age	42.5	44.5	0.003

Table 27: Misfits are younger, diverse, more educated, higher-rent, and feature less homeownership.

Type	Count	Avg Volatility	Where
"Surprise Dem"	8/30	0.807 (very high)	MI university/resort, PA Scranton
"Surprise Rep"	22/30	0.383 (moderate)	NC rural South

Table 28:"Surprise Dem" misfits are 2× more volatile

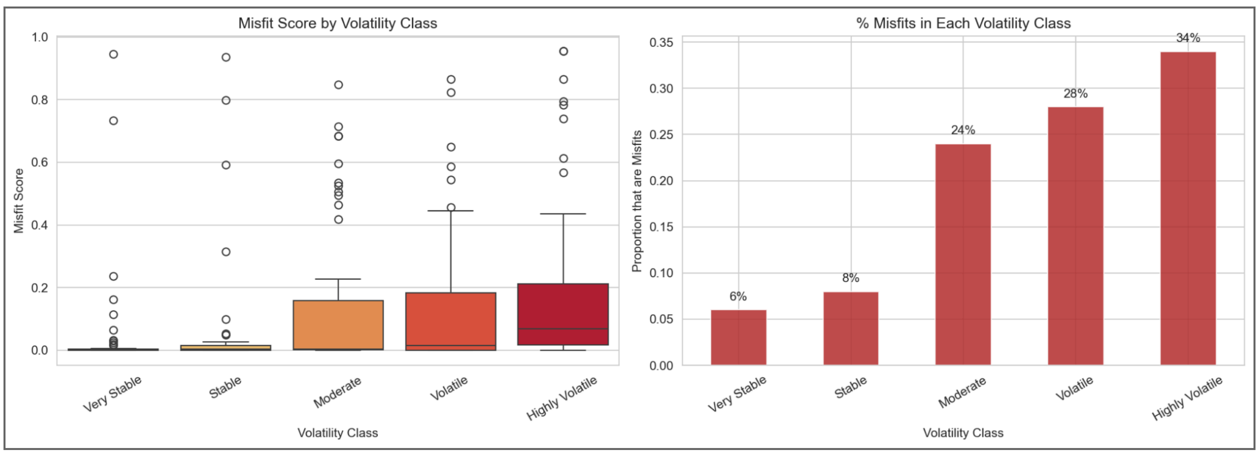

Figure 12: Misfit proportion rises from 6% (Very Stable) to 34% (Highly Volatile); Kruskal-Wallis p < 0.000001.

6.2 Investigating the Diversity Threshold

Throughout the classification and regression analyses, *race_entropy_norm* consistently emerged as one of the most influential predictors of electoral volatility. The SHAP dependence plots in Section 3.2 suggested that this relationship was not strictly linear but exhibited a threshold-like pattern, where volatility probability increased more steeply beyond a diversity level rather than rising gradually. To formalize this observation, we conducted a threshold analysis to identify the diversity value at which the probability of high volatility shifts most dramatically, and to quantify the magnitude and reliability of that shift.

The optimal threshold was identified using a maximum-separation search. We evaluated 200 candidate thresholds uniformly spaced between the 10th and 90th percentiles of the raw *race_entropy_norm* distribution. For each candidate, we computed the difference in high-volatility rates above and below the threshold. The threshold that maximized this difference was selected as the optimal split point. To assess the stability of this estimate, we performed a stratified bootstrap procedure with 1,000 resamples. Each bootstrap iteration preserved the class balance between High and Low Volatility counties and re-ran the full threshold optimization on the resampled data. The 95% confidence interval was derived from the 2.5th and 97.5th percentiles of the resulting distribution of 1,000 threshold estimates.

Metric	Value
Optimal threshold	race_entropy_norm = 0.552
95% Bootstrap CI	[0.397, 0.642]
Below: % high vol	24.4%
Above: % high vol	69.8%
Odds ratio	7.15×

Table 29: Max-separation search results for diversity threshold

The optimal threshold falls at *race_entropy_norm* = 0.552, with a 95% bootstrap confidence interval of [0.397, 0.642]**.** This threshold corresponds roughly to the point where no single racial group exceeds approximately 60% of the county population. Below the threshold, only 24.4% of counties are classified as high volatility. Above it, that rate jumps to 69.8%. The odds ratio of 7.15 indicates that counties above the diversity threshold are over seven times more likely to experience high electoral volatility than counties below it.

Figure 13 illustrates this relationship from two perspectives. The left panel overlays kernel density estimates (Gaussian KDE) of the *race_entropy_norm* distribution for Low vs High Volatility counties. The two distributions are largely separable, with Low Volatility counties concentrated below the threshold and High Volatility counties concentrated above. The gold band marks the 95% bootstrap confidence interval. The right panel plots a rolling probability curve using a 30-county sliding window, with individual counties displayed as jittered points colored by state. The curve confirms that the transition from low to high volatility probability is abrupt, rising from approximately 20% to over 70% across the threshold zone. This step-function behavior is consistent across states, as the per-state scatter points show PA, MI, and NC counties following the same pattern.

This finding reinforces the classification results from a different angle. The supervised models identified *race_entropy_norm* as the top predictor through permutation importance and SHAP analysis; the threshold analysis demonstrates that this predictive power is concentrated at a specific transition point. The implication for campaign strategists: counties approaching or crossing the diversity threshold represent the highest-priority targets for resource allocation, as they are the most likely to shift partisan lean in future election cycles.

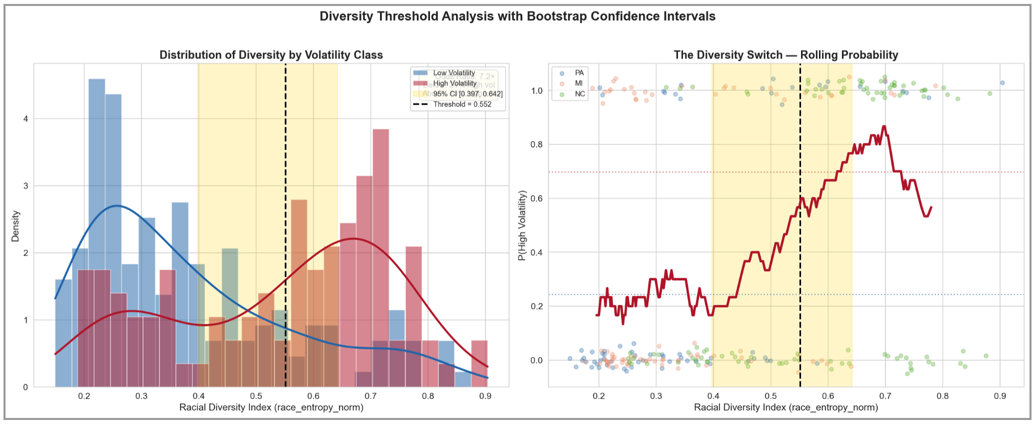

Figure 13: Diversity threshold, KDE distributions (left) and rolling probability curve (right)